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Device enrollment for server-to-device secure data exchange

Abstract

Various embodiments described herein relate to systems, methods, and non-transitory computer-readable media structured to perform server-to-device secure data exchange using a device access token. In an embodiment, a smart device receives, from a requestor entity provided to the smart device, an account data provisioning request for an account. Based on the account data provisioning request, an account identifier for the account is determined. In some arrangements, the account identifier comprises or is associated with a device access token. Based on the device access token, a data element associated with the account is determined. In some embodiments, the data element is accessible to the requestor entity only if it is not access-restricted based on the device access token. Based on the data element, an executable graphic rendering instruction is generated. The executable graphic rendering instruction is executed, which includes generating and displaying, on a user interface of the smart device, a dynamic account status indicator relating to the account.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Provisional patent application Ser. No. 17/676,328 entitled “Dynamic Account Status Indicator Via Server-To-Device Secure Data Exchange,” filed Feb. 21, 2022, which claims the benefit of and priority to U.S. Provisional Patent App. No. 63/181,861 entitled “Dynamic Account Status Indicator Via Server-To-Device Secure Data Exchange,” filed Apr. 29, 2021, and also claims the benefit of and priority to U.S. Provisional Patent App. No. 63/152,581 entitled “Server-To-Device Secure Data Exchange,” filed Feb. 23, 2021, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to server-to-device secure data exchange. More specifically, aspects of the present disclosure relate to methods, systems and computer-readable media embodying computer-executable instructions for provisioning of dynamic account status indicators via server-to-device secure data exchange. In some arrangements, the dynamic account status indicators may be related to financial accounts.

BACKGROUND

(2) Individuals use smart computing devices (e.g., smart phones, laptops, etc.) to access bank account information and perform banking activities. Individuals may also use applications provided by entities different from the bank to perform financial analytics, apply for loans, initiate automated fee disputes, etc. Such applications typically require authorization to access user data at a financial institution. Authorization typically includes a user name and password (or other credentials) provided by the user. The credentials are typically stored by the applications, which may compromise account security.

SUMMARY

(3) Various embodiments described herein relate to systems, methods, and/or non-transitory computer-readable media structured to perform server-to-device secure data exchange using a device access token. In an embodiment, a smart device receives, from a requestor entity provided to the smart device, an account data provisioning request for an account. Based on the account data provisioning request, an account identifier for the account is determined. In some arrangements, the account identifier comprises or is associated with a device access token. Based on the device access token, a data element associated with the account is determined. In some embodiments, the data element is accessible to the requestor entity only if it is not access-restricted based on the device access token. Based on the data element, an executable graphic rendering instruction is generated. The executable graphic rendering instruction is executed, which includes generating and displaying, on a user interface of the smart device, a dynamic account status indicator relating to the account.

(4) These and other features, together with the organization and manner of operation thereof, will become apparent from the following detailed description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of an example computer-implemented system structured to perform server-to-device secure data exchange, according to some arrangements;
- (2) FIG. 2 is a block diagram of an example smart device structured to facilitate server-to-device secure data exchange, according to some arrangements;
- (3) FIG. 3 is a component diagram of an example graphical user interface (GUI) of the smart device of FIG. 2, the GUI structured to facilitate smart device enrollment in server-to-device secure data exchange, according to some arrangements;
- (4) FIG. 4 is a component diagram of an example graphical user interface (GUI) of the smart device of FIG. 2, the GUI structured to allow a user to manage authorized account access settings on the smart device, according to some arrangements;
- (5) FIG. 5 is a component diagram of an example graphical user interface (GUI) of the smart device of FIG. 2, the GUI structured to allow a user to manage application settings for applications provided to the smart device, according to some arrangements;
- (6) FIG. 6 is a flowchart of an example method to facilitate smart device and/or application enrollment in server-to-device secure data exchange, according to some arrangements;
- (7) FIG. 7 is a flowchart of an example method to facilitate a transaction using the smart device in server-to-device secure data exchange, according to some arrangements;
- (8) FIG. 8 is a component diagram of an example graphical user interface (GUI) of the smart device of FIG. 2, the GUI structured to allow a user to manage application settings for use of dynamic account status indicators with account data provisioning requests by applications provided to the smart device, according to some arrangements;
- (9) FIG. 9 is a component diagram of an example graphical user interface (GUI) of the smart device of FIG. 2, the GUI structured to facilitate account data provisioning operations using

dynamic account status indicators, according to some arrangements; and (10) FIG. 10 is a flowchart of an example method to facilitate account data provisioning operations using dynamic account status indicators, according to some arrangements.

DETAILED DESCRIPTION

(11) Various embodiments described herein relate to systems, methods, and non-transitory computer-readable media structured to perform server-to-device secure data exchange using a device access token. As will be appreciated, the present disclosure provides various technical improvements and/or solves specific technical problems. For example, one of skill will recognize a technical problem of having multiple, different authentication protocols to a service provider computing system for each of the various applications provided to a smart device. Multiple authentication protocols expose both the applications and the provider computing system to security vulnerabilities, including code injection, user impersonation, interception of data in transit, interception of data at rest, etc. The present disclosure relates to one authentication protocol implemented by a component of a smart device for multiple (potentially unrelated) applications provided thereto. Furthermore, data can be provisioned to a particular smart device without being routed via computing entities that maintain the smart device and/or third-party computing applications installed on the smart device.

(12) As another example, one of skill will recognize a technical problem of allowing various applications to communicate without user intervention. The present disclosure enables a smart device to engage in secure data exchange directly with a service provider computing system by automatically managing secure authenticated sessions and provisioning data via the use of tokens, APIs, and/or SDKs.

(13) As another example, one of skill will recognize a technical problem of minimizing processing overhead and network bandwidth consumption associated with authenticated session creation and management. For example, applications that are required to provide login credentials every time they make a data request generate additional network traffic. The present disclosure enables on-demand data provisioning without requiring separate authentication each time a particular application requests data from a service provider computing system.

(14) Referring to FIG. 1, depicted is a block diagram of an example computer-implemented system **100** structured to perform server-to-device secure data exchange, according to some arrangements. In operation, the computer-implemented system **100** is structured to facilitate various secure data exchange operations (also referred to herein as “transactions”), such as data receipt, transmission, query, storage, analytics, etc.

(15) As shown, the computer-implemented system **100** includes a service provider computing system **102**, a smart device provider computing system **104**, a smart device **106**, and a third-party computing system **108**. These systems are communicatively coupled to one another via the network **113**, which enables the systems to electronically exchange data.

(16) As a general overview, the service provider computing system **102** is structured to facilitate financial services provided by a financial institution to a user of the smart device **106**. For example, the service provider computing system **102** can be managed and/or operated by a bank, credit union, insurance company, and the like. The user of the smart device **106** may have various financial accounts at the financial institution, such as a checking account, a savings account, a money market account, a mortgage or another loan account, a credit card account, etc.

(17) The smart device **106** may include any suitable electronic device, such as a smart phone, a tablet, a laptop, a desktop, a smart TV, a virtual assistant (e.g., a virtual assistant embodied as a smart speaker), an immersive reality device (e.g., a headset), a smart watch, etc. The smart device **106** may be communicatively coupled to a smart device provider computing system **104**. The smart device provider computing system **104** may be managed or operated by a manufacturer, vendor, and/or service provider that manufactures, distributes and/or services the smart device **106** (e.g., Apple®, Google®, Samsung®, etc.). The smart device provider computing system **104** may be

structured to provide software, drivers, security services, and/or other device management items to the smart device **106** in order to maintain the operation and functionality of the smart device **106**. In some arrangements, the smart device provider computing system **104** includes an app store, and a user may cause the smart device **106** to download applications therefrom. The applications may include third-party applications provided by the third-party computing system **108** (e.g., QuickBooks®, Yodlee®, etc.), as described further herein. As used herein, the term “third-party” refers to an entity that is separate and independent from the entity that operates the service provider computing system **102** and/or the smart device provider computing system **104**.

(18) In operation, the user of the smart device **106** may utilize the smart device **106** to access various services provided by the financial institution via the service provider computing system **102**, as described further herein. Further, the user of the smart device **106** may allow third-party applications associated with the third-party computing system **108** to access, via a control circuit provided to the smart device **106**, the user's account at the financial institution in order to retrieve historical data for analysis and/or aggregation and/or to perform other functions.

(19) As shown, the service provider computing system **102** includes circuitry to support server-to-device secure data exchange operations, such as token management, authenticated session management, and/or secure server-to-device data transfer. The service provider computing system **102** is shown to include various special purpose circuits, such as server token manager circuit **120**, server authentication manager circuit **122**, and server data manager circuit **124**. These circuits may retrievably store items, such as data, code, executable files, markup files, configuration files, tokens, and the like in a server secure vault **126**.

(20) The special purpose circuits (e.g., the server token manager circuit **120**, server authentication manager circuit **122**, and/or server data manager circuit **124**) may include at least one processor **130** and memory **132**. The processor **130** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a digital signal processor (DSP), a group of processing components, or other suitable electronic processing components. The memory **132** may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing and/or facilitating the various processes described herein. The memory **132** may include non-transient volatile memory, non-volatile memory, and non-transitory computer storage media. The memory **132** may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described herein. The memory **132** may be communicatively coupled to the processor **130** and include computer code or instructions for executing one or more processes described herein.

(21) The server token manager circuit **120** is structured to execute computer-based operations for managing device access tokens in a server-to-device secure data exchange ecosystem. The computer-based operations may include device enrollment management, application enrollment management, token lifecycle management, token expiration, token validation, and the like.

(22) As used herein, a “device access token” is structured to uniquely identify a particular smart device **106** to the service provider computing system **102**, which enables an enrolled smart device **106** to securely send and receive data in a server-to-device secure data exchange ecosystem. A device access token may include a device identifier, a financial account identifier, a user identifier for the smart device provider computing system **104**, an application identifier, a timestamp, and/or other elements sufficient to authenticate a device and/or an application. One or more elements of each of the device identifier, financial account identifier, user identifier, and/or application identifier may be included. A device identifier may include a model number, a serial number, a Wi-fi address (e.g., MAC address, Bluetooth device address), an international mobile equipment identifier (IMEI), a mobile equipment identifier (MEID), an integrated circuit card identifier (e.g., subscriber identification module (SIM) card identifier), etc. A financial account identifier may

include a hash of a financial account number or an otherwise obscured financial account number in whole or in part. A user identifier may include, for example, a social networking handle, an e-mail address, a phone number, a user's user name for the service provider computing system **102**, a user's user name for the smart device provider computing system **104**, and/or a user's user name for the third-party computing system **108**. An application identifier may include, in whole or in part, an application name, an application instance identifier, an installation and/or last update timestamp for a particular application instance, etc. In some arrangements, these items may be converted to a string and concatenated to form a device access token. In some arrangements, the device access token is a mark-up language file (e.g., an XML file) where each particular data element is identified by a unique tag. In some arrangements, the device access token is a quick response (QR) code or another machine-readable optical label displayable via a display screen of the smart device **106** (e.g., for troubleshooting, for sharing with secondary smart device(s) **106**, etc.).

(23) The server authentication manager circuit **122** is structured to execute computer-based operations for authenticating smart devices **106** in a server-to-device secure data exchange ecosystem. The server authentication manager circuit **122** may receive, from a particular smart device **106**, an electronic message that includes a request for data access and a device access token. The server authentication manager circuit **122** may parse one or more device identifiers from the device access token. The server authentication manager circuit **122** may cross-reference the one or more device identifiers to the device identifier(s) previously stored in the server secure vault **126** in order to determine whether a particular device has been previously enrolled. The server authentication manager circuit **122** may compare various parsed items from the device access token to verify that a particular enrolled smart device **106** is not being spoofed (e.g., impersonated) by an unauthorized device. For example, the server authentication manager circuit **122** may parse a SIM card identifier and a MAC address or a Bluetooth device address from a device access token and determine that a device is unauthorized if a known SIM card identifier is accompanied by a new MAC address or a Bluetooth device address, likely indicative of the SIM card having been removed from a previously authorized smart device **106** and installed on a different device. In another example, the server authentication manager circuit **122** may parse, from the device access token or from Internet traffic information associated with the request (e.g., from a header, footer, payload, or metadata properties of the packets of data received at the service provider computing system **102** in connection with the request for data), the source network identifier and compare the identifier to a list of previously stored known access networks for a particular smart device **106**. The network identifier can include an IP address, a subnet, a service set identifier (SSID) for a wireless network, or another suitable identifier. In yet another example, the service provider computing system **102** may receive a geographical location identifier (e.g., a set of coordinates) from the smart device **106** and may compare the geographical location identifier to a set of previously known locations for the smart device **106**.

(24) As part of authenticating a particular smart device **106**, the server authentication manager circuit **122** may also receive (e.g., as part of a device access token, as a separate element in an electronic message, or in a separate electronic message) an application identifier for an application provided to the smart device **106**. As used herein, the term "provided to" refers to an application that includes functionality accessible to a user via the smart device **106**. In some arrangements, the application is installed on the smart device **106**. In some arrangements, the application is executing on the smart device **106** (e.g., via a browser). In some arrangements, the application is accessible at the smart device **106** via an emulator or a similar application delivery framework (e.g., Citrix®, Azure®, etc.), and is installed on and/or executing on a remote computing system relative to the smart device **106**. A particular application may have an associated set of access permissions and/or restrictions that allow the application to perform certain specific functions and/or access specific data provided by the service provider computing system **102**. The server authentication manager circuit **122** may cross-reference the application identifier to the application identifier(s) and the

corresponding restriction(s) previously stored in the server secure vault **126** in order to determine whether a particular application provided to the smart device **106** has been previously enrolled and before retrieving and transmitting the requested data back to the smart device **106** for use by the application.

(25) As part of authenticating a particular smart device **106**, the server authentication manager circuit **122** may perform lifecycle-related checks on the received device access token. For instance, the server authentication manager circuit **122** may access a timestamp (e.g., a token creation time, a token expiration time, a token last used time) previously stored in the server secure vault **126** to determine if the received device access token is valid and/or if the user of the smart device **106** needs to complete an additional authentication process. For example, if a predetermined amount of time (e.g., one day, seven days, thirty days, never before used, etc.) has passed since a particular device access token was last used, the server authentication manager circuit may generate and cause the smart device **106** to provide to the user (e.g., in a display form, in an audible form) a prompt requesting the user's login credentials, biometric information, and/or authorization to proceed.

(26) As part of authenticating a particular smart device **106**, the server authentication manager circuit **122** may work in concert with the smart device **106** to manage secure authorized sessions. A secure authorized session may establish time boundaries for processing a particular data request from an authenticated device. For example, the server authentication manager circuit **122** may receive, together or separately from the device access token, a session identifier for a secure authorized session established by the smart device **106** for the purpose of data transmission. If the server authentication manager circuit **122** is unable to validate the device access token and/or application, the server authentication manager circuit **122** may transmit an electronic message to the smart device **106**. The electronic message may include the session identifier and instructions to the smart device **106** to terminate the secure authorized session. The authentication manager circuit **122** may also be structured to receive electronic messages from the smart device **106** indicating that a particular secure authenticated session has been terminated, in which case the requested data will not be transmitted to the smart device **106**.

(27) The server data manager circuit **124** is structured to execute computer-based operations for data provisioning to smart devices **106** in a server-to-device secure data exchange ecosystem. After a device access token is verified and as long as a secure authenticated session between the service provider computing system **102** and the smart device **106** is active, the server data manager circuit **124** may be structured to retrieve and/or provide the data requested by a particular smart device **106**. The server data manager circuit **124** may also execute the requested functionality, such as initiate a dispute, request a fee waiver, disable a particular card, etc. Based on the application identifier, the service provider computing system **102** may also access and apply application-specific restrictions in the server secure vault **126**, as discussed further herein.

(28) To carry out its operations, server data manager circuit **124** may be structured to determine a financial account identifier of the user. In some arrangements, the server data manager circuit **124** may parse the financial account identifier from the data request message received from the smart device **106**. In some arrangements, the server data manager circuit **124** may parse the financial account identifier from the device access token received from the smart device **106**. To improve security of user data, the financial account identifier may be encoded for provisioning and storage by the smart device **106** by, for example, generating a hash of a financial account number or otherwise obscuring the financial account identifier in whole or in part. The server data manager circuit **124** may apply a decoding algorithm and/or cross-reference the received encoded financial account identifier to a list previously stored in the server secure vault **126** in order to determine the actual account identifier. In some arrangements, the server data manager circuit **124** may receive a user identifier for the smart device provider computing system **104** (e.g., Apple® ID, Google® user name, Samsung® ID, etc.) and determine the actual account identifier(s) for the user's financial

account(s) based on the user identifier for the smart device provider computing system **104**.

(29) The service provider computing system **102** is communicatively coupled to the smart device provider computing system **104**, smart device **106**, and third-party computing system **108** via network **113**. To communicate via the network **113**, the service provider computing system **102** includes a network circuit **134**. The network circuit **134** may be used to establish connections with other computing devices by way of the network **113**. The network circuit **134** may include program logic that facilitates connection of the service provider computing system **102** to the network **113**. In some arrangements, the network circuit **134** may include any combination of a wireless network transceiver (e.g., a cellular modem, a Bluetooth transceiver, a Wi-Fi transceiver) and/or a wired network transceiver (e.g., an Ethernet transceiver). For example, the network circuit **134** may include an Ethernet device such as an Ethernet card and machine-readable media such as an Ethernet driver configured to facilitate connections with the network **113**. In some arrangements, the network circuit **134** includes the hardware and machine-readable media sufficient to support communication over multiple channels of data communication. Further, in some arrangements, the network circuit **134** includes cryptography capabilities to establish a secure or relatively secure communication session in which data communicated over the session is encrypted.

(30) Although not shown in FIG. **1**, it is understood that device provider computing system **104**, smart device **106**, and/or third-party computing system **108** may include network interfaces for long-, medium- or short-range communication substantially similar to the network circuit **134** as described above.

(31) The network **113** may include a local area network (LAN), a wide area network (WAN), a telephone network, such as the Public Switched Telephone Network (PSTN), a wireless link, an intranet, the Internet, or combinations thereof. The network **113** can enable communication between various nodes. In some arrangements, data flows through the network **113** from a source node to a destination node as a flow of data packets, e.g., in the form of data packets in accordance with the Open Systems Interconnection (OSI) layers. A flow of packets may use, for example, an OSI layer-4 transport protocol such as the User Datagram Protocol (UDP), the Transmission Control Protocol (TCP), or the Stream Control Transmission Protocol (SCTP), transmitted via the network **113** layered over an OSI layer-3 network protocol such as Internet Protocol (IP), e.g., IPv4 or IPv6. The network **113** is composed of various network devices (nodes) communicatively linked to form one or more data communication paths between participating devices. Each networked device includes at least one network interface for receiving and/or transmitting data, typically as one or more data packets. An illustrative network **113** is the Internet; however, other networks may be used. The network **113** may be an autonomous system (AS), i.e., a network that is operated under a consistent unified routing policy (or at least appears to from outside the AS network) and is generally managed by a single administrative entity (e.g., a system operator, administrator, or administrative group).

(32) The network **113** may be composed of multiple connected sub-networks or AS networks, which may meet at one or more of: an intervening network (a transit network), a dual-homed gateway node, a point of presence (POP), an Internet eXchange Point (IXP), and/or additional other network boundaries. The network **113** can be a local-area network (LAN) such as a company intranet, a metropolitan area network (MAN), a wide area network (WAN), an inter network such as the Internet, or a peer-to-peer network, e.g., an ad hoc Wi-Fi peer-to-peer network. The data links between nodes in the network **113** may be any combination of physical links (e.g., fiber optic, mesh, coaxial, twisted-pair such as Cat-5 or Cat-6, etc.) and/or wireless links (e.g., radio, satellite, microwave, etc.).

(33) The network **113** can include carrier networks for mobile communication devices, e.g., networks implementing wireless communication protocols such as the Global System for Mobile Communications (GSM), Code Division Multiple Access (CDMA), Time Division Synchronous Code Division Multiple Access (TD-SCDMA), Long-Term Evolution (LTE), or any other such

protocol including so-called generation 3G, 4G, 5G, and 6G protocols. The network **113** can include short-range wireless links, e.g., via Wi-Fi, BLUETOOTH, BLE, or ZIGBEE, sometimes referred to as a personal area network (PAN) or mesh network. The network **113** may be public, private, or a combination of public and private networks. The network **113** may be any type and/or form of data network and/or communication network.

(34) The network **113** can include a network interface controller that can manage data exchanges with devices in the network **113** via a network interface (sometimes referred to as a network interface port). The network interface controller handles the physical and data link layers of the Open Systems Interconnection (OSI) model for network communication. In some arrangements, some of the network interface controller's tasks are handled by one or more processing circuits. In various arrangements, the network interface controller is incorporated into the one or more processing circuits, e.g., as circuitry on the same chip.

(35) In some arrangements, the network interface controller supports wireless network connections and an interface is a wireless (e.g., radio) receiver/transmitter (e.g., for any of the IEEE 802.11 Wi-Fi protocols, near field communication (NFC), BLUETOOTH, BLUETOOTH LOW ENERGY (BLE), ZIGBEE, ANT, or any other wireless protocol). In various arrangements, the network interface controller implements one or more network protocols such as Ethernet.

(36) Referring now to FIG. 2, depicted is a block diagram **200** of an example smart device **106** structured to facilitate server-to-device secure data exchange, according to some arrangements. In operation, the smart device **106** is structured to facilitate various secure data exchange operations, such as data receipt, transmission, query, storage, analytics, etc.

(37) As shown in a simplified view, the smart device **106** includes hardware **201**, operating system **202**, and applications **208**. The operating system **202** is shown to include a kernel **203**, a core services circuit **204**, and a control circuit **206**. The control circuit **206** can be a special purpose circuit structured to facilitate server-to-device secure data exchange between the smart device **106** and/or a secondary smart device **240** and the service provider computing system **102** of FIG. 1. It is understood that the control circuit **206** may be implemented, in whole or in part, as part of the operating system **202** and/or as one or more of the applications **208**. Further, the control circuit **206** may be structured to include various hardware **201** components described further herein.

(38) As shown, hardware **201** includes a processor **210**, memory **212**, a secure element **214**, an input/output (I/O) circuit **216**, a communications circuit **218**, and a sensor **219**. The processor **210** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a digital signal processor (DSP), a group of processing components, or other suitable electronic processing components. The memory **212** may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing and/or facilitating the various processes described herein. The memory **212** may include non-transient volatile memory, non-volatile memory, and non-transitory computer storage media. The memory **212** may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described herein. The memory **212** may be communicatively coupled to the processor **210** and include computer code or instructions for executing one or more processes described herein.

(39) In some arrangements, the memory **212** is included in, at least in part, or is communicatively coupled to the secure element **214**. The secure element **214** can be a removable or built-in hardware and/or software circuit structured to securely store data and/or securely host applications **208** on the smart device **106**. Further, in some arrangements, the secure element **214** may store executables for invocation by the various circuitry included in the core services circuit **204**, control circuit **206**, and/or applications **208**. Further, in some arrangements, the secure element **214** may include a dedicated or shared memory space for execution of these various processes (e.g., by the kernel **203** and/or by the control circuit **206**). The secure element can be implemented as an embedded

computer chip, a removable SIM card, a system-on-a-chip (SoC), or similar. In some arrangements, the secure element **214** includes a co-processor additional to the processor **210**. In some arrangements, the secure element **214** includes or is communicatively coupled to a near-field communications (NFC) controller, such as the communications circuit **218**. More generally, the communications circuit **218** may include a transceiver suitable for short-, medium- or long-range communication, such as a wireless network transceiver (e.g., a cellular modem, a Bluetooth transceiver, a Wi-Fi transceiver, an NFC transceiver, etc.).

(40) The I/O circuit **216** includes suitable input/output ports and/or uses an interconnect bus for interconnection with a local display (e.g., a liquid crystal display, a touchscreen display) and/or keyboard/mouse devices (when applicable), or the like, serving as a local user interface for programming and/or data entry, retrieval, or other user interaction purposes. As such, the I/O circuit **216** may provide an interface for the user to interact with various applications **208**. For example, the I/O circuit **216** may include a keyboard, a keypad, a mouse, joystick, a touch screen, a microphone, a biometric device (e.g., a fingerprint sensor), a virtual reality headset, smart glasses, and the like. As another example, the I/O circuit **216** may include, but is not limited to, a television monitor, a computer monitor, a speaker, and so on. In some arrangements, the I/O circuit **216** includes a camera suitable for taking photographic images and/or scanning QR codes using the smart device **106**.

(41) The sensor **219** may include circuitry and/or a transceiver suitable for collecting and/or outputting various data. For example, the sensor **219** may be a global positioning system (GPS) transceiver configured to detect a geographical location (e.g., latitude and longitude) of smart device **106** in real or near-real time by using triangulation based on the coordinates of one or more cellular towers received by the smart device **106** via the communications circuit **218**.

(42) As shown, the operating system **202** includes a kernel **203**, a core services circuit **204**, and a control circuit **206**. The kernel **203** is structured to work in conjunction with the core services circuit **204**. Accordingly, the kernel **203** can include a dedicated space in the memory **212** for executing various processes managed by the core services circuit **204**. These processes can include, for example, process management, file management, networking, user interface management, driver management for connected devices, and the like. The executables for these and similar services may be stored in the memory **212** and invoked, monitored, and terminated by the core services circuit **204**. In some arrangements, the kernel **203** and/or the core services circuit **204** may include kernel extension executables for the control circuit **206** (i.e. the control circuit **206** may be included in the kernel **203** and/or the core services circuit **204** at least in part.)

(43) The control circuit **206** is a special purpose circuit structured to facilitate server-to-device secure data exchange operations. In some arrangements, the control circuit **206** may receive requests and/or provide data to the applications **208**. As shown, the applications **208** can include a service provider application **230**, a device-native application **232**, and a third-party application **234**. The service provider application **230** may be, for example, a mobile banking application structured to exchange data with the service provider computing system **102**. The service provider application **230** may include various functionality, such as account lookup, balance lookup, transaction history lookup, etc. for a financial account of a user. Accordingly, the information related to the financial account of the user may be accessible via the smart device **106**. The device-native application **232** may be developed and/or provided to the smart device **106** by an operator of the smart device provider computing system **104**, and may include an Internet browser, a camera control application, a telephone control application, an app store application, a control application for the secondary smart device **240**, and the like. The third-party application **234** may be developed and/or provided by an operator of the third-party computing system **108**. The third-party application may be independently downloaded by a user of the smart device **106** or may be provided via an app store application managed by the smart device provider computing system **104** (e.g., the device-native application **232**). The third-party application **234** may be configured to access the user's account at

the financial institution in order to retrieve historical data for analysis and/or aggregation and/or to perform other functions (e.g., automated fee disputes, underwriting, etc.).

(44) Any of the applications **208** may be configured, via the control circuit **206**, to access the user's financial data at the financial institution associated with the service provider computing system **102**. As such, the technical problem of enhancing data security is solved by device-based authentication such that the control circuit **206** can bypass the smart device provider computing system **104** in providing data from the smart device **106** to applications **208** not managed by the financial institution (e.g., in providing data to the device-native application **232** and/or the third-party application **234**) and/or to the secondary smart device **240**. Furthermore, the server-to-device secure data exchange infrastructure managed by the control circuit **206** in concert with the service provider computing system **102** allows for minimization or significant reduction of the amount of private data (e.g., personally identifiable information (PII)) stored on the smart device **106**. Furthermore, tokenization of confidential account information prevents the applications **208** from locally accessing and/or storing account identifiers of a user. As described further herein, account restrictions may further define the type of data, functionality, and/or data uses allowable for each application **208**.

(45) As shown, the control circuit **206** includes a device token manager circuit **220**, a device authentication manager circuit **222**, an authorized session manager circuit **223**, a device data manager circuit **224**, and a device secure vault **226**. In operation, the control circuit **206** works in concert with the service provider computing system **102** to facilitate the enrollment of the smart device **106**, secondary smart device **240**, and/or particular applications **208** in the server-to-device secure data exchange ecosystem. Further, the control circuit **206** allows a user of the smart device **106** to provide authorized account access settings for the applications **208**. Further, the control circuit **206** allows the applications **208** and/or the secondary smart device **240** to initiate transactions (e.g., data downloads, queries, funds transfer requests, etc.) from the smart device **106**.

(46) The device token manager circuit **220** is structured to execute computer-based operations for managing device access tokens in a server-to-device secure data exchange ecosystem. The computer-based operations may include device enrollment management, application enrollment management, token lifecycle management, token expiration, token validation, and the like, as described relative to FIG. 3-5.

(47) The device authentication manager circuit **222** is structured to execute computer-based operations for authenticating smart devices **106** in a server-to-device secure data exchange ecosystem. The device authentication manager circuit **222** may receive, via a GUI rendered on the smart device **106**, a smart device identifier.

(48) In some arrangements, the smart device **106** is the device associated with the smart device identifier, and the user or an application **208** attempts to access and receive data at the smart device **106**. In some arrangements, a first smart device **106** (e.g., a mobile device, a tablet, a laptop, a desktop, etc.) is a full-functionality device that includes the functionality of the control circuit **206** sufficient to perform the functions described herein. A secondary smart device **240** (e.g., a virtual assistant, a smart watch, an immersive reality device) may be designated by a user as an authorized device for receiving at least some of the data provided via server-to-device secure data exchange. The secondary smart device **240** may be associated with an application **208** (e.g., a fob issued by a financial institution may be associated with the service provider application **230**, a virtual assistant device may be associated with a device-native application **232** that controls the virtual assistant device, and/or an internet-of-things device, such as a smart home component, may be associated with a third-party application **234**, etc.). In this case, the device authentication manager circuit **222** may receive, at the smart device **106**, a secondary device identifier related to the secondary smart device **240**, and, upon prompting a user to approve a proposed secure data exchange transaction, may retrieve a corresponding token stored in the device secure vault **226** on a secure element **214**. In some arrangements, the requesting application **208** is also identified, and the application

identifier may be separately retrieved or may be included in a particular device access token.

(49) After receiving a device identifier, the device authentication manager circuit **222** may generate and transmit the device access token and/or the application identifier to the server authentication manager circuit **122** of the service provider computing device **102**, which may retrieve and provide the requested data. According to various embodiments, the device authentication manager circuit **222** and/or the server authentication manager circuit **122** may apply the relevant restrictions prior to providing the data to the requesting computing device and/or application **208**.

(50) As part of authenticating a particular smart device **106** or secondary smart device **240** for a particular data request, the authorized session manager circuit **223** may initiate and manage a secure authorized session (e.g., a secure time-limited communications session between the smart device **106** and the service provider computing device **102**). The authorized session manager circuit **223** may generate a session identifier for a secure authorized session established by the smart device **106** for the purpose of data transmission between a requestor device (e.g., smart device **106**) and/or requestor application **208** and the service provider computing device **102**. In some arrangements, the secure authorized session is established after validating the request at the smart device **106** (e.g., after verifying that a device access token exists and is not expired, and that the applicable access restrictions are met). In some arrangements, only some or none of the foregoing operations are performed at the smart device **106**, such that the server authentication manager circuit **122** performs further token validation, as described above, and causes the authorized session manager circuit **223** to terminate the secure communications session if the appropriate server-side checks performed by the service provider computing system **102** have failed.

(51) As part of authenticating a particular smart device **106** or secondary smart device **240** for a particular data request, the authorized session manager circuit **223** may terminate a particular secure authorized session according to predetermined criteria. For example, a secure authorized session may be terminated at the smart device **106** if no response is received from the service provider computing system **102** within a predetermined amount of time (e.g., 15 sec., 30 sec., etc.), if the size of an inbound data transmission exceeds a predetermined threshold (e.g., 5 MB, 10 MB, etc.), if a user device enters inactive or shutdown mode, if a code injection attempt is detected as described below, etc.

(52) The device data manager circuit **224** is structured to execute computer-based operations for data requests to the service provider computing system **102** from requestor application(s) **208** at the smart device **106**. The device data manager circuit **224** may provide (e.g., access, retrieve from memory **212** and/or secure element **214**) an API and/or SDK comprising executables for invocation by requestor application(s) **208**. The executable(s) may be selectively tagged (e.g., in a configuration file implemented as a mark-up language file, such as XML, and stored in memory **212** and/or on secure element **214**) with permission labels corresponding to restrictions.

Accordingly, the device data manager circuit **224** may provide to the requestor application(s) **208** only the allowable executables for permissible (non-restricted) function calls. In some arrangements, the application(s) **208** include the appropriate parameters for the executables (e.g., “retrieve.exe” parametrized with an account identifier, amount(s), date range(s) for transactions to retrieve, etc.). In some arrangements, the device data manager circuit **224** receives the parameter arguments from the application(s) **208** and constructs the parametrized function calls in order to prevent errors in execution and minimize the possibilities for code injection. If a valid command is not detected or cannot be constructed, the device data manager circuit **224** may cause the authorized session manager circuit **223** to terminate the corresponding secure authenticated session.

(53) The device data manager circuit **224** is structured to execute computer-based operations for data provisioning to requestor application(s) **208** at the smart device **106**. The device data manager circuit **224** may receive the requested data from the service provider computing device, and may make the data available to the requestor application(s) **208**. In some arrangements, when a requestor application **208** is an intermediary for processing data requests from the secondary smart

device **240**, the device data manager circuit **224** may transmit the requested data set directly to the secondary smart device **240**, which may perform post-processing of the received data thereon and/or provide the results to the user.

(54) Referring now to FIG. **3**, depicted is a component diagram of an example graphical user interface (GUI) **300** of the smart device **106** of FIG. **2**, the GUI structured to facilitate smart device **106** enrollment in server-to-device secure data exchange, according to some arrangements. As shown, a user may provide login credentials (e.g., a user name **302** and a password **304**) for the service provider computing system **102** via a data input control (e.g., a text box) of the GUI and then actuate the log in control **306** (e.g., a button). Upon receiving the login credentials, the service provider computing system **102** may transmit an electronic message to the smart device **106**, causing the smart device **106** to generate and display an enroll smart device user interface **310**. The enroll smart device user interface **310** provides a front-end to allow the user to interact with the control circuit **206** of the smart device **106**. For example, a list of smart devices **312** may be generated and provided to the user. The list of smart devices **312** may include smart devices associated with the user, either previously enrolled or known to be associated with the user (e.g., by determining the secondary smart devices **240** locally paired to the smart device **106** via Bluetooth or similar, by scanning a QR code provided by a particular secondary smart device **240**, by receiving and decoding an NFC token from a particular secondary smart device **240** at the smart device **106**, etc.).

(55) For each selected smart device **106** or secondary smart device **240** in the list of smart devices **312**, the user can use the select accounts control **314** to specify financial accounts to which the selected smart device **106** or secondary smart device **240** should have access. Upon detecting a user interaction with a generate token control **316**, the device token manager circuit **220** of the smart device **106** may generate a device access token **320** for the selected device and/or selected account. An example device access token **320** may include one or more device identifiers **322**, financial account identifiers **324**, and/or financial account type (checking, savings, credit card, etc.) identifiers **326**, as shown. The device access token may be stored by the device token manager circuit **220** in device secure vault **226**, which may be stored in the memory **212** and/or secure element **214** of the smart device **106**.

(56) Referring now to FIG. **4**, depicted is a component diagram of an example graphical user interface (GUI) **330** of the smart device **106** of FIG. **2**, the GUI structured to allow a user to manage authorized account access settings via the smart device **106**, according to some arrangements. The GUI **330** is structured to provide to a user a list of account-level restrictions **334** selectable and configurable to fine-tune the level of granularity in account access. For example, the user may utilize account restriction controls **336** to specify operations that are allowable for the control circuit **206** of the smart device **106** to initiate for users and/or third-party applications. The account restriction controls **336** may include, for example, whether a physical and/or virtual card associated with a particular account can be turned on/off (e.g., activated/deactivated for financial and/or non-financial transactions) using server-to-device authentication, whether specific transaction data can be received using server-to-device authentication, whether disputes can be automatically initiated using server-to-device authentication, etc. The application restriction controls **338** further allow users to apply different account restriction controls **336** to specific applications. For example, a user may allow a greater scope of functionality to the service provider application **230** relative to the third-party application **234**. In another example, a user may allow a higher level of functionality if a particular application **208** is a trusted application. In some arrangements, the restrictions can be stored and applied locally on the smart device **106** before initiating a secure authorized session with the service provider computing system **102**. In some arrangements, the restrictions can be stored and applied by the service provider computing system **102**. In some arrangements, the restrictions can be included in device access tokens **320** and parsed from the device access tokens **320** before being applied. In some arrangements, the restrictions can

be stored in a markup-language file and applied to select and/or parametrize only allowable function calls from an API or SDK library, which may be retrievably stored in the device secure vault **226** of the smart device **106**, in the server secure vault **126** of the provider computing system **102**, or in another suitable location.

(57) Referring now to FIG. 5, depicted is a component diagram of an example graphical user interface (GUI) **340** of the smart device **106** of FIG. 2, the GUI structured to allow a user to manage application settings for applications **208** provided to the smart device **106**, according to some arrangements.

(58) As shown, the user can further restrict various applications **208** provided to the smart device **106** by defining specific restrictions **348** (access and/or data use levels) for each combination of a particular application **344** and account restriction controls **346**. For example, the user may specify that a particular application **344** can receive transaction data (an example account restriction control **346**) only when the sensor **219** (e.g., a GPS sensor) provides data to the smart device **106** and/or to the service provider computing system **102** that indicates that a user is within a predetermined radius of a predetermined geographical location, within a particular geofenced area, etc. In another example, the user may specify that a particular application **344** can receive transaction data (an example account restriction control **346**) only for transactions that meet a particular threshold, fall in a particular date range, have a particular transaction descriptor, etc. In another example, the user may specify that a particular application **344** can receive transaction data (an example account restriction control **346**) only for specific uses. For example, in some arrangements, an application **208** may be restricted from storing a data set, may be allowed to receive only summary data, may be allowed to receive only de-identified data that excludes PII, etc.

(59) Referring now to FIG. 6, depicted is a flowchart of an example method **350** to facilitate smart device **106** and/or application **208** enrollment in server-to-device secure data exchange, according to some arrangements. As a brief overview, the method **350** includes operations to enroll a smart device **106** and/or application **208** to securely send and receive data in a server-to-device secure data exchange ecosystem. The application **208** may be any of a service provider application **230**, a device-native application **232**, and/or a third-party application **234**.

(60) At **352**, a smart device selection is received via a user interface provided on a display screen of the smart device **106**. In some arrangements, the smart device selection refers to the smart device **106** (i.e., when the smart device **106** is initially enrolled in server-to-device data exchange). In some arrangements, the smart device selection refers to a secondary smart device **240**, which the user is configuring for enrollment in the server-to-device data exchange ecosystem via a previously enrolled smart device **106**. At **354**, an additional selection of a specific application **208** may be received at the smart device **106**. At **356**, an additional selection of a specific financial account may be received at the smart device **106**. At **358**, a device access token is generated at the smart device **106** and/or at the service provider computing system **102**. The device access token may include a device identifier, a financial account identifier, a user identifier for the smart device provider computing system **104**, an application identifier, a timestamp, and/or other elements sufficient to authenticate a device and/or an application. At **360**, the device access token is retrievably stored. In some arrangements, the device access token is retrievably stored in the device secure vault **226**, which may be included in the memory **212** and/or secure element **214** of the smart device **106**. In some arrangements, the device access token is retrievably stored in the server secure vault **126** of the service provider computing system **102**. At **362**, various user selections related to authorized account access settings and/or account restrictions, as described above, are received at the smart device **106**. As described above, the authorized account access settings may be stored on the smart device **106** and/or on the service provider computing system **102**.

(61) Referring now to FIG. 7, depicted is a flowchart of an example method **370** to facilitate a transaction using the smart device **106** and/or a secondary smart device **240** in server-to-device

secure data exchange, according to some arrangements. As a brief overview, the method **370** includes operations to allow a smart device **106** and/or the secondary smart device **240** to perform secure data exchange using a previously stored device access token. Any of the previously enrolled applications **208** (a service provider application **230**, a device-native application **232**, and/or a third-party application **234**) may engage in server-to-device secure data exchange, subject to the appropriate restrictions set using the control circuit **206** of the smart device **106**.

(62) At **372**, a secure authorized session between the smart device **106** and the service provider computing system is established as described relative to FIGS. **1** and **2**. These operations may be performed contemporaneously or sequentially, in any suitable order, relative to receiving a transaction request from a particular application **208**. For example, in some arrangements, an authorized secure session is automatically created for an active application **208** before said application **208** generates a request for a transaction. In some arrangements, an authorized secure session is created after the transaction request is received at the smart device **106** and/or after the transaction request is validated, at **376**, at the smart device **106** and/or at the service provider computing system **102**.

(63) As part of validating the transaction request at **376**, the smart device **106** may identify a requestor application **208** using an application name, an application instance identifier, an installation and/or last update timestamp for a particular application instance, etc. The smart device **106** may retrieve a previously stored device access token corresponding to the application **208** and verify that the token is valid (e.g., the token is not expired, the token was created after the installation and/or last update timestamp for the application **208**, etc.). The smart device **106** may further retrieve previously stored restrictions associated with the application **208** and/or the device access token and apply the restrictions to the request and/or transmit the restrictions to the service provider computing system **102** for application of the restrictions. Applying the restrictions may include, for example, scrubbing (validating) and/or constructing allowable function calls using an API or SDK for performing the relevant transaction (e.g., for accessing the relevant data, for invoking the relevant functionality, etc.) Applying the restrictions may further include determining, based on the request, a specific account identifier associated with the transaction and validating that the account is on a list of specific accounts or subaccounts for which the requested transaction is allowed.

(64) At **380**, the smart device **106** receives an electronic response message from the service provider computing system **102**. The response message may include a data set generated by the service provider computing system **102** in response to the request. At **382**, the received data set is provided to the requestor application **208**.

(65) Referring now to FIG. **8**, depicted is a component diagram of an example graphical user interface (GUI) **340** of the smart device **106** of FIG. **2**. The GUI **340** is structured to allow a user to manage application settings **802** for use of dynamic account status indicators **816** with account data provisioning requests. As a general overview, dynamic account status indicators **816** may be associated with financial accounts and structured to enable differentiated dynamic presentation of account attributes to the user of the smart device **106** via the GUI **340**. The differentiated dynamic presentation systems and methods described herein solve a technical problem of customizing account presentation based on confidential account-related data without providing the confidential account-related data to the requestor entity and/or by ensuring that the requestor entity has authorization to access the confidential account-related data on a particular smart device **106**.

According to various embodiments, the requestor entity may be a digital wallet application. In such embodiments, the dynamic account status indicators may cause a graphical representation of a payment account (e.g., a payment card image) to change (e.g., change color or design, change an opacity, change a shape, etc.) or have information or graphics overlaid on top of the representation (e.g., text overlaid on top of the payment card image).

(66) Accordingly, one or more applications provided to the smart device **106** may include digital

wallet features. For example, any of the service provider application **230**, device-native application **232**, or third party application **234** may be structured to allow the user of the smart device **106** to generate, manage, and/or use digital identities for the user's real-world financial accounts (e.g., a checking account, a savings account, a brokerage account, a credit card account, a rewards account, etc.). In some arrangements, such as when the service provider application **230** is a digital wallet application, the entity that provides, manages, or administers the financial account also provides, manages, or administers the digital wallet application. In some arrangements, such as when the device-native application **232** and/or third party application **234** is a digital wallet application, the entity that provides, manages, or administers the financial account is different from the entity that provides, manages, or administers the digital wallet application. For example, the operator of the device-native application **232** may also provide a pre-installed or downloadable device-native digital wallet application. In some arrangements, the device-native digital wallet application may be downloadable from the smart device provider computing system **104** via an app store provided by the operator of the smart device provider computing system **104**. In another arrangement, the digital wallet application may be provided and/or downloadable from the third-party computing system **108**.

(67) The digital identities of the financial accounts may include various account attributes, such as account identifiers, account numbers, PIN code(s), login credentials, expiration dates, current balances, transaction history, credit limits, remaining available spend, daily cash withdrawal limits, daily purchase limits, card enabled/disabled indicators, reward points usage conditions, rewards points balances, geographical restrictions on use, etc. In some embodiments, the account attributes are replacement values used to obscure actual values, and a cross-reference structure that maps these values may be stored on or off the smart device **106**. In some embodiments, the cross-reference structure may include one or more device access tokens generated for the account. In some embodiments, the account attributes are tokenized values generated, for example, as hashes of the respective actual values according to suitable hashing algorithms. In some embodiments, at least some attributes of the digital identities are stored in the memory **212** and/or on the secure element **214** of the smart device **106**. In some embodiments, at least some attributes of the digital identities are stored remotely relative to the smart device **106** for example, on data storage media associated with the service provider computing system **102**, such as the server secure vault **126**.

(68) Various attributes of digital identities may be used to enable differentiated dynamic presentation of account attributes to the user of the smart device **106** via the GUI **340**. In an example use case, a quick glance indicator (also sometimes referred to as a dynamic account status indicator) may be associated with a digital identity of a financial account and rendered via the GUI **340**. According to various embodiments, the quick glance indicator may comprise a card image or another graphics- and/or text-based informational representation entity rendered via the GUI **340**. In some embodiments, the quick glance indicator may include one or more properties associated with account data and/or with the card image or another informational representation entity. The property may include a particular value or range of values for color and/or opacity such that, for example, when an account balance reaches a predetermined threshold, the color and/or opacity of the informational representation entity are set to a specified value. The property may include a particular value or range of values for location-specific use. For example, a particular card may be enabled or disabled for use in specified geographical locations, at specified merchants, etc. The property may further include a binary enabled/disabled indicator.

(69) In an example arrangement of FIG. **8**, the dynamic account status indicator settings **804**, identified by the header **802**, can be pre-defined by the service provider computing system **102**. The control circuit **206** of the smart device **106** may retrieve the settings from the server secure vault **126** via a query, API call, SDK function call, or another suitable electronic data request message. The dynamic account status indicator settings **804** may be retrievably stored relative to identifier(s) or data regarding a particular requestor entity (e.g., relative to an application instance

identifier for a digital wallet application).

(70) The dynamic account status indicator settings **804** may be further editable by the user of the smart device **106**. For example, as shown, the user may specify how various account data items (account type, balance, credit limit, daily limit, transaction data, rewards data, etc.) should be used to generate the quick glance indicator. For each item shown at **804**, the user may further modify or define the criteria **810**, such as the amount **812**, percentage **814**, indicator value **816**, card status **818**, etc. In an example use case, the user may, for example, specify that if an account balance is at or above the amount **812** (e.g., \$4,800), the indicator value **816** is set to “red” and the card status **818** is set to “off” (e.g., disabled) such that the card cannot be used for further purchases. As a result, the quick glance indicator for the corresponding digital identity may include a displayable card image that is colored red and has an “X” over it to indicate that the card is not available for use, as shown in FIG. 9. The user may use the next control **820** to define another set of parameters—for example, if the account balance is at or above \$3,000, the indicator value **816** is set to “yellow”, if the account balance is at or below \$1,000, the indicator value **816** is set to “green”, etc. In some embodiments, the threshold amounts are evaluated relative to the credit limit if the account is a credit account. In some embodiments, the threshold amounts are evaluated relative to the available balance of funds if the account is a deposit account. Further, in various embodiments, percentage **814** thresholds defined relative to the daily limit, account balance, or the credit limit can be used instead or in addition to amount **812** thresholds. In some embodiments, the GUI **340** allows the user to navigate to an account management website or application provided by the service provider computing system **102**. Accordingly, the user may define or modify the settings shown according to FIG. 8 via the banking website or application.

(71) One of skill in the art will appreciate that the term “card”, as used herein, may be used interchangeably with the term “account”. Accordingly, the digital wallet application may include a digital identity for a virtual account that does not have a corresponding physical card.

(72) In operation, the quick glance indicator is dynamically generated when a particular requestor entity provides a request for account data to the smart device **106**. For example, the smart device **106** may receive an account data provisioning request from a digital wallet application when the user of the smart device **106** opens and/or activates the digital wallet application or attempts to perform a transaction. The request may include an actual or obscured account identifier. For example, the requestor entity may retrieve and include in the request a previously stored account identifier, which may be part of a digital identity for an account. The smart device **106** may determine a device access token based on the requestor entity and/or the account identifier. In some embodiments, the smart device **106** may parse the request into one or more data items (e.g., account balance, credit limit), verify, based on the device access token, that these data items are not access-restricted for the requestor entity, access retrievably stored dynamic account status indicator settings **804**, retrieve the relevant data items, apply the dynamic account status indicator settings **804** to the data items, and generate a displayable quick glance indicator. The displayable quick glance indicator may be rendered on the smart device **106**. In some embodiments, the data items are not persisted (not stored in non-volatile memory) on the smart device **106**, which improves data security. Furthermore, once the displayable quick glance indicator is generated and rendered in response to a particular data request, the smart device **106** may be structured to clear its cache or otherwise discard the data from its transitory memory.

(73) Referring now to FIG. 9, depicted is a component diagram of an example graphical user interface (GUI) **340** of the smart device **106** of FIG. 2. The GUI **340** is structured to facilitate account data provisioning operations using dynamic account status indicators, according to some arrangements. In some arrangements, the user may navigate to the GUI **340** from the display screen shown in FIG. 8. In some arrangements, the user may navigate to the GUI **340** by accessing the digital wallet application on the smart device **106**. In some arrangements, the user may navigate to the GUI **340** when performing a financial transaction, for example, responsive to bringing the smart

device **106** in proximity to a merchant's or another funds recipient's device, or more generally, the third-party computing system **108**, and activating an NFC communications interface, a Bluetooth communications interface, or another suitable communications interface.

(74) As shown according to an example embodiment, the GUI **340** is structured to include one or more informational representation entities that correspond to one or more digital identities of the user's accounts. In some embodiments, the one or more digital identities are accessible via a digital wallet application. As shown, in the example arrangement, the user of the smart device **106** has three accounts, each represented by a respective card image **910**, **920**, or **930**. As shown according to an example embodiment, each of the card images **910**, **920**, or **930** is rendered as an image of a payment card; however, any suitable displayable entity comprising an image, text, and/or computer-executable instructions (e.g., navigation controls, data access controls, etc.) can be used.

(75) As shown, an example card image **910** is associated with a dynamic account status indicator **919** and a navigable control structured to retrieve and display further information **912** and/or **932**. In some embodiments, the dynamic account status indicator **919** may define a set of properties for the card image **910**. The set of properties may be populated when the dynamic account status indicator **919** is generated. The dynamic account status indicator **919** may be programmatically bound to the card image **910** as an attribute, as a navigable reference (e.g., hyperlink) to a markup-language file (e.g., an XML file), or using another suitable method. In some embodiments, the dynamic account status indicator **919** is a record set retrieved responsive to a query, an API function call, and/or an SDK function call, or via another suitable electronic messaging interface. In some embodiments, the dynamic account status indicator **919** may exist only in volatile memory of the smart device **106**.

(76) In operation according to an example use case, the user accesses a digital wallet application on the smart device **106** via the GUI **340**. The user is presented with a first card image **910** for a first account, a second card image **920** for a second account, and a third card image **930** for a third account. The user may wish to use one or more of the respective accounts to perform a funds transfer transaction (e.g., to make a purchase at a store).

(77) As shown, the first account is disabled and unavailable for use. The user may tap, click or otherwise interact with the first card image **910** to activate a navigation control. The navigation control may be structured to provide further information **912** regarding the data items that caused the dynamic status indicator **919** to be generated for the first account. For example, as shown according to the dynamic status indicator **919**, the user may have used the user interface of FIG. 8 to disable the account and set the color of the card image to red when the balance exceeds \$4,800 relative to the credit limit. As shown according to further information **912**, the user's current account balance **912** is \$4,900 relative to the credit limit of \$5,000. As shown, in some embodiments, the user may interact with the disable card control **918** to make the account available for use despite the restrictions.

(78) To generate and render the first card image **910**, when the user accesses the GUI **340** via the digital wallet application, the digital wallet application, also sometimes referred to as a requestor entity, generates a request for data sufficient to generate and render the first card image **910** for the user's account. The smart device **106** retrieves, from the secure element **214** or memory **212**, a device access token based on the requestor entity and/or the account identifier included in the request. The smart device **106** accesses retrievably stored dynamic account status indicator settings (defined, for example, as described in reference to FIG. 8) for the user's account. The smart device **106** retrieves the relevant data items (here, the account balance **912** and account limit **914**) and applies the dynamic account status indicator settings to the data items. Applying the settings to data items may include various suitable operations, such as value comparisons to thresholds, enabling or disabling accounts for use, enabling or disabling geographical restrictions, etc. Accordingly, the smart device **106** generates the dynamic status indicator **919**.

(79) Further, the smart device **106** may implement access controls for the data items, which may

include confidential account-related information. For example, in some embodiments, the smart device **106** may verify, based on the device access token and according to the control settings described in relation to FIGS. **3-5**, that the data items, such as the balance **914** and credit limit **916**, are not access-restricted for the requestor entity. If not access-restricted, the smart device **106** may provide further information in an electronic response message directly to the requestor entity (e.g., the digital wallet application). If items are access-restricted, the smart device **106** may generate and display a pop-up message not accessible or modifiable by the requestor entity to show further information **912**. Accordingly, in some embodiments, the requestor entity may be provided by the smart device **106** only with the dynamic status indicator **919**, which may be sufficient to define the appearance of the card image **910** without revealing confidential account-related information.

(80) As shown, the second account is enabled and available for use. The second account, corresponding to the second card image **920**, may be selected by the user to complete the transaction. In some embodiments, the second account is automatically selected as a payment method for the transaction if other accounts are unavailable for use or are in a “yellow” or “red” state. More generally, the digital wallet application may be structured to automatically select, as a payment method, an account that is the best candidate for use relative to other accounts of the user (e.g., “green” vs. “yellow”, “yellow” vs. “red”, etc.).

(81) As shown, the third account is disabled and unavailable for use. The user may tap, click or otherwise interact with the third card image **930** to activate a navigation control. The navigation control may be structured to provide further information **932**. Here, in an example use case, further information **932** may include a digital map **934**. The digital map **934** may include an indication of Merchant A location **934a** and the user's current location **934b**. The user's current location may correspond to the current location of the smart device **106** determined using a GPS transceiver or another type of sensor **219** associated with the smart device **106**. The dynamic status indicator generated for the third account may include a location use restriction that prevents the third account from being used at Merchant A (e.g., if the smart device **106** is within a predetermined distance from Merchant A). Accordingly, further information **932** may include a notification **936** explaining why the third account is unavailable for use. Further information **932** may also include a disable card control **938**, and, in some embodiments, the user may interact with the disable card control **918** to make the account available for use despite the restrictions.

(82) Referring now to FIG. **10**, depicted is a flowchart of an example method **1000** to facilitate account data provisioning operations using dynamic account status indicators, according to some arrangements. As a brief overview, the method **1000** includes operations to enable a smart device **106** to facilitate the generation and rendering, on a user interface of the smart device **106**, of account information according to a dynamic account status indicator. Example embodiments of dynamic account status indicators are described in relation to FIGS. **8** and **9**. The operations of method **1000** may be performed by the service provider computing system **102**, smart device provider computing system **104**, smart device **106**, and/or third-party computing system **108**.

(83) At **1002**, the smart device **106** receives an account data provisioning request. The account data provisioning request may be generated by any of the service provider application **230**, device-native application **232**, or third-party application **234** provided to the smart device **106**. The account data provisioning request may include an account identifier. In some embodiments, the account identifier is associated with a device access token. The device access token is unique to the smart device and/or a combination of a smart device, particular financial account of a user, and application instance identifier for the requestor entity (or another suitable identifier for the requestor entity, such as its associated computing system **102**, **104** or **108**). In some embodiments, the account identifier is a device access token. In some embodiments, the account identifier is an actual account identifier value of the financial account. In some embodiments, the account identifier obscures or replaces the actual account identifier. In some embodiments, the requestor entity is a digital wallet application and the account identifier is associated with the application and

retrievably stored in the memory **212** of the smart device **106** and/or in memory associated with the requestor computing system, such as any of the systems **102**, **104**, or **106**. In some embodiments, the account identifier is received by the requestor entity via a function call from an API or SDK library made available by the service provider computing system **102** to the requestor entity.

(84) At **1004**, the smart device **106** (e.g., the control circuit **206**) may extract the account identifier from the electronic request message and, based on the extracted account identifier, determine the corresponding device access token. To determine the device access token, the smart device **106** may also extract from the electronic request message an application instance identifier for the application that generated the function call, or another suitable identifier for the requestor entity. An example device access token **320** is described in relation to FIG. **3**. The device access token may be stored in the memory **212** and/or secure element **214** of the smart device **106**. In some embodiments, the smart device **106** may tokenize and/or detokenize the account identifier and/or application instance identifier received from the requestor entity to generate values in a format consistent with that of the device access token. The smart device **106** may then retrieve the retrievably stored device access token, based on the received values, from the memory **212** and/or secure element **214**.

(85) At **1006**, smart device **106** (e.g., the control circuit **206**) may identify an access-controlled data element sufficient to generate a dynamic account status indicator. The smart device **106** may access (e.g., in the memory **212**, secure element **214**, and/or via a function call to the service provider computing system **102**) the dynamic account status indicator settings **804**, which may include the criteria **810** as described in relation to FIG. **8**. The smart device **102** may determine, based on the retrieved settings and/or criteria, which confidential data elements associated with a user's financial account are needed to generate the dynamic account status indicator. For example, if the settings and/or criteria call for displaying a graphical representation of the account in a particular color based on the account balance, the corresponding confidential data elements may include the account balance.

(86) The smart device **102** may determine, based on the device access token and the control restrictions described relative to FIGS. **3-5**, whether the requestor entity is authorized to receive the confidential data element. If the requestor entity is authorized to receive the confidential data element, the smart device **106** may provide the data element to the requestor entity, and the operations **1008** and **1010** may be performed by the requestor entity at least in part. If the requestor entity is not authorized to receive the data element, the operations **1008** and **1010** may be performed by the control circuit **206**. In some embodiments, however, the control circuit **206** may be structured to delegate some aspects of the operations **1008** and/or **1010** to the requestor entity if these aspects do not involve receiving confidential data by the requestor entity. For example, the control circuit **206** may generate parametrized function calls for rendering card images and provide the same to the requestor entity for execution on the smart device **106**.

(87) At **1008**, the smart device **106** (e.g., the control circuit **206**) may generate an executable graphic rendering instruction. In some embodiments, the executable graphic rendering instruction is a function call structured to generate the definitions for user interface elements of FIG. **9**, such as the card image **910**. As described in relation to FIG. **9**, the card image **910** may have a dynamic account status indicator **919** associated therewith. The smart device **106** may populate the dynamic account status indicator **919** properties or data items with appropriate values determined based on the retrieved settings and/or criteria. For example, if the account balance indicates that a card should be shown in red (or otherwise marked to visually, audibly, or haptically indicate a restriction or a lack of a restriction such as, for example, sounding a particular tone or combination of tones corresponding to a particular alert level) and disabled for use, the smart device **106** may populate the relevant dynamic account status indicator **919** properties with appropriate values. Accordingly, the output of operations **1008** may include an executable instruction (e.g., an .exe file). The executable instruction may comprise a static or dynamic reference to a previously stored image file

(e.g., a card image file) and the dynamic account status indicator **919** properties that customize the appearance and/or functionality of the previously stored image file. In some arrangements, the smart device **106** provides the executable instruction to the requestor entity for execution at **1010**. As will be appreciated, in such arrangements, the executable instruction may not include any confidential account information but rather includes the dynamic account status indicator **919**. In some arrangements, however, the smart device **106** may retrieve (e.g., via a function call to the service provider computing system **102**) previously stored further information regarding the account and may provide the same to the requestor entity.

(88) At **1010**, the requestor entity generates and displays the requested card image modified according to the dynamic account status indicator **919**. In some embodiments, the requestor entity executes the executable instruction received at **1008**. In some embodiments, the executable instruction is a response to the function call made by the requestor entity at **1002**. The output of the operations at **1010** may be a graphical user interface rendered on the smart device **106** and comprising at least one card image, such as the card image **910** shown in FIG. 9.

(89) The embodiments described herein have been described with reference to drawings. The drawings illustrate certain details of specific embodiments that provide the systems, methods and programs described herein. However, describing the embodiments with drawings should not be construed as imposing on the disclosure any limitations that may be present in the drawings.

(90) It should be understood that no claim element herein is to be construed under the provisions of 35 U.S.C. § 112(f), unless the element is expressly recited using the phrase “means for.”

(91) As used herein, the term “circuit” may include hardware structured to execute the functions described herein. In some embodiments, each respective “circuit” may include machine-readable media for configuring the hardware to execute the functions described herein. The circuit may be embodied as one or more circuitry components including, but not limited to, processing circuitry, network interfaces, peripheral devices, input devices, output devices, sensors, etc. In some embodiments, a circuit may take the form of one or more analog circuits, electronic circuits (e.g., integrated circuits (IC), discrete circuits, system on a chip (SOCs) circuits, etc.), telecommunication circuits, hybrid circuits, and any other type of “circuit.” In this regard, the “circuit” may include any type of component for accomplishing or facilitating achievement of the operations described herein. For example, a circuit as described herein may include one or more transistors, logic gates (e.g., NAND, AND, NOR, OR, XOR, NOT, XNOR, etc.), resistors, multiplexers, registers, capacitors, inductors, diodes, wiring, and so on).

(92) The “circuit” may also include one or more processors communicatively coupled to one or more memory or memory devices. In this regard, the one or more processors may execute instructions stored in the memory or may execute instructions otherwise accessible to the one or more processors. In some embodiments, the one or more processors may be embodied in various ways. The one or more processors may be constructed in a manner sufficient to perform at least the operations described herein. In some embodiments, the one or more processors may be shared by multiple circuits (e.g., circuit A and circuit B may comprise or otherwise share the same processor which, in some example embodiments, may execute instructions stored, or otherwise accessed, via different areas of memory).

(93) Alternatively or additionally, the one or more processors may be structured to perform or otherwise execute certain operations independent of one or more co-processors. In other example embodiments, two or more processors may be coupled via a bus to enable independent, parallel, pipelined, or multi-threaded instruction execution. Each processor may be provided as one or more general-purpose processors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), digital signal processors (DSPs), or other suitable electronic data processing components structured to execute instructions provided by memory. The one or more processors may take the form of a single core processor, multi-core processor (e.g., a dual core processor, triple core processor, quad core processor, etc.), microprocessor, etc. In some embodiments, the one

or more processors may be external to the apparatus, for example the one or more processors may be a remote processor (e.g., a cloud based processor). Alternatively or additionally, the one or more processors may be internal and/or local to the apparatus. In this regard, a given circuit or components thereof may be disposed locally (e.g., as part of a local server, a local computing system, etc.) or remotely (e.g., as part of a remote server such as a cloud based server). To that end, a “circuit” as described herein may include components that are distributed across one or more locations.

(94) An exemplary system for providing the overall system or portions of the embodiments might include a general purpose computing computers in the form of computers, including a processing unit, a system memory, and a system bus that couples various system components including the system memory to the processing unit. Each memory device may include non-transient volatile storage media, non-volatile storage media, non-transitory storage media (e.g., one or more volatile and/or non-volatile memories), etc. In some embodiments, the non-volatile media may take the form of ROM, flash memory (e.g., flash memory such as NAND, 6 NAND, NOR, 6 NOR, etc.), EEPROM, MRAM, magnetic storage, hard discs, optical discs, etc. In other embodiments, the volatile storage media may take the form of RAM, TRAM, ZRAM, etc. Combinations of the above are also included within the scope of machine-readable media. In this regard, machine-executable instructions comprise, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions. Each respective memory device may be operable to maintain or otherwise store information relating to the operations performed by one or more associated circuits, including processor instructions and related data (e.g., database components, object code components, script components, etc.), in accordance with the example embodiments described herein.

(95) It should also be noted that the term “input devices,” as described herein, may include any type of input device including, but not limited to, a keyboard, a keypad, a mouse, joystick or other input devices performing a similar function. Comparatively, the term “output device,” as described herein, may include any type of output device including, but not limited to, a computer monitor, printer, facsimile machine, or other output devices performing a similar function.

(96) Any foregoing references to currency or funds are intended to include fiat currencies, non-fiat currencies (e.g., precious metals), and math-based currencies (often referred to as cryptocurrencies). Examples of math-based currencies include Bitcoin, Litecoin, Dogecoin, and the like.

(97) It should be noted that although the diagrams herein may show a specific order and composition of method steps, it is understood that the order of these steps may differ from what is depicted. For example, two or more steps may be performed concurrently or with partial concurrence. Also, some method steps that are performed as discrete steps may be combined, steps being performed as a combined step may be separated into discrete steps, the sequence of certain processes may be reversed or otherwise varied, and the nature or number of discrete processes may be altered or varied. The order or sequence of any element or apparatus may be varied or substituted according to alternative embodiments. Accordingly, all such modifications are intended to be included within the scope of the present disclosure as defined in the appended claims. Such variations will depend on the machine-readable media and hardware systems chosen and on designer choice. It is understood that all such variations are within the scope of the disclosure. Likewise, software and web implementations of the present disclosure may be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various database searching steps, correlation steps, comparison steps and decision steps.

(98) The foregoing description of embodiments has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from this disclosure. The embodiments were chosen and described in order to explain the

principals of the disclosure and its practical application to enable one skilled in the art to utilize the various embodiments and with various modifications as are suited to the particular use contemplated. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and arrangement of the embodiments without departing from the scope of the present disclosure as expressed in the appended claims.

Claims

1. A method comprising: receiving, by a smart device via a first software application executing on the smart device, a request to enroll the smart device in a server-to-device secure data exchange ecosystem that allows unrelated applications executing on the smart device to transact with a computing system of a first service provider indirectly via the smart device; receiving, by the smart device, a first selection of a financial account held by the first service provider, and a second selection of a second software application of a second service provider distinct from the first service provider; responsive to receiving the first selection and the second selection generating, by the smart device, a device access token based on (i) a device identifier corresponding to the smart device, (ii) a financial account identifier corresponding to the financial account, (iii) a user identifier corresponding to a user of the smart device, and (iv) an identifier of the second software application; retrievably storing, by the smart device in a secure storage element of the smart device, the device access token in association with the second software application; receiving, by the smart device, a transaction request from the second software application executing on the smart device; responsive to receiving the transaction request, (i) establishing, by the smart device, a secure authorized session between the smart device and the computing system of the first service provider, and (ii) accessing, by the smart device, the previously stored device access token according to the identifier of the second software application; validating, by the smart device, the device access token with respect to the second software application; transmitting, by the smart device to the computing system, via the secure authorized session, the device access token and one of (i) the transaction request, or (ii) a modified transaction request, wherein the computing system of the first service provider determines that the transaction request is valid by parsing the identifier of the second software application encoded as part of the device access token; receiving, by the smart device from the computing system, via the secure authorized session, an electronic message responsive to the transaction request or to the modified transaction request, the electronic message provided based on the device access token being validated; and providing, by the smart device to the second software application, a response to the transaction request based on the electronic message.
2. The method of claim 1, wherein the first selection and the second selection are received via the first software application executing on the smart device.
3. The method of claim 1, further comprising receiving, by the smart device, an account restriction applicable to the second software application.
4. The method of claim 3, wherein the account restriction is received via the first software application.
5. The method of claim 3, further comprising storing the account restriction in the secure storage element in association with the second software application.
6. The method of claim 3, further comprising generating, by the smart device, the modified transaction request based on the transaction request and the account restriction.
7. The method of claim 6, wherein the modified transaction request is transmitted to the computing system via the secure authorized session.
8. The method of claim 6, wherein generating the modified transaction request comprises applying the account restriction to the transaction request.
9. The method of claim 3, further comprising transmitting the account restriction to the computing

system via the secure authorized session.

10. The method of claim 1, wherein the transaction request is a request for data.

11. A smart device comprising one or more processors configured to: receive, via a first software application executing on the smart device, a request to enroll the smart device in a server-to-device secure data exchange ecosystem that allows unrelated applications executing on the smart device to transact with a computing system of a first service provider indirectly via the smart device; receive a first selection of a financial account held by the first service provider, and a second selection of a second software application of a second service provider distinct from the first service provider; responsive to receiving the first selection and the second selection generate a device access token based on (i) a device identifier corresponding to the smart device, (ii) a financial account identifier corresponding to the financial account, (iii) a user identifier corresponding to a user of the smart device, and (iv) an identifier of the second software application; retrievably store, in a secure storage element of the smart device, the device access token in association with the second software application; receive a transaction request from the second software application executing on the smart device; responsive to receiving the transaction request, (i) establish a secure authorized session between the smart device and the computing system of the first service provider, and (ii) access the previously stored device access token according to the identifier of the second software application; validate the previously stored device access token with respect to the second software application; transmit, via the secure authorized session, the device access token and one of (i) the transaction request, or (ii) a modified transaction request, wherein the computing system of the first service provider determines that the transaction request is valid by parsing the identifier of the second software application encoded as part of the device access token; receive, from the computing system, via the secure authorized session, an electronic message responsive to the transaction request or to the modified transaction request, the electronic message provided based on the device access token being validated; and provide, to the second software application, a response to the transaction request based on the electronic message.

12. The smart device of claim 11, wherein the first selection and the second selection are received via the first software application executing on the smart device.

13. The smart device of claim 11, the one or more processors further configured to receive an account restriction applicable to the second software application.

14. The smart device of claim 13, wherein the one or more processors are configured to receive the account restriction via the first software application.

15. The smart device of claim 13, the one or more processors further configured to store the account restriction in the secure storage element in association with the second software application.

16. The smart device of claim 13, the one or more processors further configured to generate the modified transaction request based on the transaction request and the account restriction.

17. The smart device of claim 16, wherein the modified transaction request is transmitted to the computing system via the secure authorized session.

18. The smart device of claim 16, wherein generating the modified transaction request comprises applying, by the one or more processors, the account restriction to the transaction request.

19. The smart device of claim 13, the one or more processors further configured to transmit the account restriction to the computing system via the secure authorized session.

20. The smart device of claim 11, wherein the response comprises the electronic message received from the computing system.
